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CERTIFICATIONS

Machine Learning Specialization
Coursera
Certified Mar 2023

Google Data Analytics Professional Certificate
Coursera
Certified Apr 2025

Linux Essentials Certificate
Cisco
Certified Dec 2024

Six Sigma White Belt Certificate
Educatе 360
Certified May 2024

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C, C++, R, SQL, Bash/Shell scripting, and MATLAB.

Web Development: MongoDB, Express, React, Node.js, REST APIs, WebSocket, HTML, CSS, and Bootstrap.

Data Analysis & Visualization: Power BI, Pandas, NumPy, and Matplotlib.

Machine Learning & AI: scikit-learn, TensorFlow, and PyTorch.

Tools & Technologies: Git, Jupyter Notebooks, NetBeans, Docker and Eclipse.

Operating System: Windows, macOS, Linux, Ubuntu, and Unix.

Hardware: Full Assembly Code (HCS12), Verilog, FPGAs, MIPS Assembly, Microcontrollers, and Embedded Systems

Ryan Attia

EDUCATION

Masters in Computer Science - Georgia Tech University
May 2026 - Present (Expected Graduation Dec 2027)

Graduate Studies in Artificial Intelligence - Pennsylvania State University
Aug 2025 - May 2026

B.S. in Computer Engineering - Pennsylvania State University
Aug 2020 - May 2024

PROJECTS

Diabetic Foot Ulcer Risk Chatbot

Jan 2026 – Present

- Preprocessed 100,000+ patient records from the Diabetes 130-US Hospitals dataset, engineering clinical features across age, glucose levels, A1C results, medication usage, and diagnosis history to train a TensorFlow/Keras feedforward neural network for binary DFU risk classification.
- Built an end-to-end chatbot intake pipeline that guides patients through a structured questionnaire, validates and normalizes inputs, encodes categorical medical variables, and feeds a model-ready feature vector to generate a real-time DFU probability score and risk classification.
- Designed preprocessing logic to align user responses with the exact feature order expected at inference, serializing model artifacts to ensure consistent and reproducible predictions across sessions.
- **Skills:** Python, TensorFlow/Keras, Deep Learning, Neural Networks, Scikit-learn, Conversational AI, Healthcare AI, Feature Engineering, Data Preprocessing, Risk Classification.

AI-Powered Pac-Man

Nov 2025 - April 2026

- Building an AI-controlled Pac-Man game in Python using Pygame, featuring maps with multiple interactive elements across the map throughout the multiple levels.
- Implementing a Q-learning reinforcement learning agent to optimize path selection and reward collection.
- Developing adaptive enemy agents using Breadth-First Search, constantly learning to provide ~70% success rate on easy, ~50% on medium, and ~30% on hard difficulty.
- Optimizing game loop for seamless animations and near-zero input delay for responsive agent control.
- **Skills:** Python, Pygame, Reinforcement Learning, Q-Learning, BFS Pathfinding, Game AI, State Management, Real-Time Systems, Algorithm Design.

Group Chat Application

Aug 2024 – March 2026

- Built a web-based chat application enabling real-time communication through a responsive React interface with conversation search, group chats, and live user status.
- Designed a Node.js + Express backend, developing a REST API for authentication with JWT-based security.
- Integrated MongoDB to manage over 10K+ user and message records using a scalable NoSQL schema.
- Implemented Socket.IO with WebSockets to achieve instant message delivery (<300ms latency) and enhance user engagement through event-driven interactions.
- **Skills:** MERN Stack, JWT, WebSocket, REST APIs, React, JavaScript, and Mongoose.

Object Classification Tool

Jan - March 2024

- Developed an 18-layer Convolutional Neural Network in Python for object detection, classifying 32×32×3 images into 10 categories.
- Implemented six types of layers in the neural network for image normalization, convolution, ReLU, Maxpool, fully-connected, and Softmax, ensuring precise implementation and performance.
- Applied the network to a dataset of 10,000 images, fine-tuning the network in response to classification errors to enhance accuracy and reliability.
- **Skills:** MATLAB, Convolutional Neural Networks, Image Classification, Deep Learning, Data Preprocessing.

WORK EXPERIENCE

Java Backend Developer Intern – Revature

Jan - Oct 2025

- Built and optimized RESTful backend services using Java and Spring (Core, Boot, JDBC) to support enterprise applications handling 1K–2K daily API requests.
- Collaborated with frontend teams to integrate APIs, implement JWT-based authentication, and reduce average response time through efficient query handling and caching.
- Engaged in Agile sprints, code reviews, and unit testing, improving system reliability across development environments.
- Utilized Git, SQL, and Spring MVC best practices to ensure modular and production-ready backend components.
- **Skills:** Java, Spring Boot, Spring Core, JDBC, REST APIs, SQL, Git, JWT, MVC Architecture, Agile, CI/CD

Retail Sales Representative – Unilever

July 2024 – Jan 2026

- Collaborated with retail management to execute merchandising strategies that enhanced product visibility and aligned with Unilever's brand goals to drive growth.
- Drove a 50% increase in territorial revenue compared to the previous year across all Unilever product categories through effective sales execution, relationship management, and market optimization.

Software Engineer – WSSC

Jan – May 2024

- Designed a multilingual interactive display system (Programmable Water Leak Demonstration Unit) that educates users on the impact of household water leaks using a real-time leak detection solution.
- Developed the backend in Python to process sensor data and calculate water flow in gallons per minute, achieving 99% leak detection accuracy and enabling timely identification of leaks.
- Built the frontend in Java, designing an interface for real-time data visualization that improved accessibility and user engagement for WSSC's educational demonstrations.
- **Skills:** Python, Java, Raspberry Pi, real-time data processing, sensor integration, and GUI development.